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Majorana corner modes and tunable braiding in an altermagnet–superconductor heterostructure

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Zeeman fields are usually required for realizing and manipulating Majorana corner modes at the cost of suppressed superconducting pairing. Altermagnetism offers a new route toward realizing Majorana corner modes under zero net external magnetic fields. Here we propose a theoretical scheme to host Majorana corner modes in a honeycomb-lattice altermagnet–superconductor heterostructure. In contrast to previous studies on square lattices, altermagnetism yields distinct effective low-energy edge masses on various inequivalent edges of the honeycomb lattice. Within a low-energy edge-theory framework, we establish the conditions under which Majorana corner modes emerge for both bearded and zigzag edge terminations. We further explicitly construct a complete set of Clifford gates by tuning the altermagnetic anisotropy and chemical potential of the heterostructure. Our results provide a promising platform for the realization and manipulation of Majorana corner modes.

I. INTRODUCTION AND MOTIVATION

Majorana zero modes (MZMs) serve as building blocks for topological qubits, and their coherent control enables the implementation of topological quantum gates [1, 2]. The pursuit and manipulation of MZMs therefore constitute a central objective in topological quantum computation. Early theoretical proposals for MZM realization were rooted in two-dimensional p-wave superconductors and the one-dimensional Kitaev chain [3, 4]. Nevertheless, intrinsic p-wave pairing remains challenging to achieve in both condensed-matter systems and ultracold atomic gases. To engineer effective p-wave pairing, numerous theoretical schemes have been put forward leveraging the synergy between spin-orbit coupling and s-wave superconductor pairing with Zeeman fields [5–8]. Substantial progress has been made in the theoretical description and control of Majorana qubits [9–14]. These topological-superconductor platforms generally require Zeeman fields to break time-reversal symmetry, and braiding operations for MZMs in two-dimensional systems typically rely on manipulating magnetically induced vortices.

Higher-order topological superconductors offer an alternative platform for constructing topological quantum gates, as they host Majorana modes at lower-dimensional boundaries [15–27]. For instance, second-order topological superconductors support Majorana zero modes localized at sample corners [28–30]. Braiding of these Majorana corner modes can therefore be achieved via parameter tuning, without the need to manipulate superconducting vortices [31–35]. Nevertheless, existing schemes for generating isolated Majorana corner modes still typically rely on Zeeman fields. Taken together, realizations and manipulations of two-dimensional Majorana zero modes have so far generally required magnetic fields, which tend to suppress superconducting pairing.

Altermagnets exhibit collinear compensated magnetic order with momentum-dependent exchange fields [36–38]. Their vanishing net magnetic field avoids Pauli pair breaking of superconductivity [39]. Altermagnetism breaks time-reversal symmetry. Such symmetry breaking provides a prerequisite for isolated nondegenerate Majorana corner modes for topological quantum computation. Studies on square-lattice systems show the interplay between d-wave altermagnetism and s-wave superconductivity can host Majorana corner modes [40, 41]. These proposals exploit anisotropic effective mass on orthogonal edges originating from altermagnetic order. Other altermagnetic heterostructures also provide routes to Majorana zero modes [42–44]. Majorana braiding has also been numerically simulated in patterned altermagnetic platforms [45]. Variations in lattice symmetry modify mass projections along edges for other lattice geometries. The honeycomb lattice supports zigzag, armchair, and bearded edges. Studies on honeycomb lattices have identified higher-order insulating phases in the Kane-Mele model [46] and Majorana corner modes in altermagnet-superconductor heterostructures [47]. However, the mechanism for altermagnetism-induced edge mass projections and Majorana braiding on honeycomb lattices remains unclear.

In this work, we show that Majorana corner modes can arise at corners formed by bearded or zigzag edges in the d-wave altermagnet–superconductor heterostructure. The altermagnetic edge-mass projections depend on both edge orientation and termination, leading to distinct conditions for the emergence of corner modes. Tuning the altermagnetic anisotropy and chemical potential then allows their positions to be adiabatically controlled. This enables corner-pair selection, paired transport in a hexagon, and exchange of a single pair in a triangle. We further extend this edge-mass control scheme to coupled-flake networks and construct the X, Hadamard, and CNOT gates within fixed-parity encodings.

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II. MODEL HAMILTONIAN FOR THE PHYSICAL SYSTEM

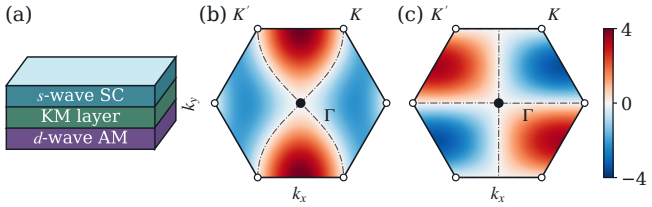


FIG. 1. (a) Illustration of a Kane–Mele layer sandwiched between an s-wave superconductor and a d-wave altermagnet. (b)–(c) Altermagnetic form factor $f_{\rm AM}(\mathbf{k},\chi)/\lambda$ for $\chi=0$ ($d_{x^2-y^2}$) and $\chi=\pi/4$ (d_{xy}), respectively. The hexagon denotes the first Brillouin zone. Dash-dotted lines mark the altermagnetic nodes. Both panels adopt the same color scale.

Consider a Kane-Mele layer [48] coupled to an s-wave superconductor and a d-wave altermagnet as shown in Fig. 1(a). The Hamiltonian for the system reads

$$H = -t\sum_{\langle ij\rangle} c_i^\dagger c_j + it'\sum_{\langle\langle ij\rangle\rangle}\nu_{ij}c_i^\dagger s_z c_j + \sum_i(\varepsilon_i-\mu)c_i^\dagger c_i + \lambda\sum_{\langle\langle ij\rangle\rangle}\eta_{ij}(\chi)c_i^\dagger(\boldsymbol{s}\cdot\hat{\boldsymbol{n}})c_j + \sum_i\left(\Delta_s c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger + \text{H.c.}\right), \tag{1}$$

where $c_i=(c_{i\uparrow},c_{i\downarrow})^T$ denotes the annihilation operator spinor for spin-up and spin-down fermions at site i. $\langle ij\rangle$ and $\langle\langle ij\rangle\rangle$ label nearest-neighbor and next-nearest-neighbor site pairs, respectively. t and t' denote the nearest-neighbor hopping, intrinsic spin–orbit coupling, respectively. The chirality factor for the intrinsic spin–orbit coupling is $\nu_{ij}=\text{sgn}[\hat{\boldsymbol{z}}\cdot(\hat{\boldsymbol{d}}_{ik}\times\hat{\boldsymbol{d}}_{jk})]=\pm1$.The unit vectors $\hat{\boldsymbol{d}}_{ik}$ and $\hat{\boldsymbol{d}}_{jk}$ point from the intermediate lattice site k to sites i and j. s_α with $\alpha=x,y,z$ denote the Pauli matrices acting on the spin degrees of freedom. $\varepsilon_i=M\xi_i$ defines a staggered potential with $\xi_i=1$ on the A sublattice and $\xi_i=-1$ on the B sublattice, respectively. λ is the magnitude of the d-wave altermagnetic order, and $\eta_{ij}(\chi)=\cos[2(\theta_{ij}-\chi)]$ encodes the d-wave anisotropy with $\theta_{ij}=\arg(\boldsymbol{r}_i-\boldsymbol{r}_j)$. Here $\boldsymbol{r}_i$ and $\boldsymbol{r}_j$ are the position vectors of sites i and j, respectively. The unit vector $\hat{\boldsymbol{n}}$ specifies the orientation of the Néel vector. In-plane Néel orientations are connected by global spin rotations about the $\hat{\boldsymbol{z}}$ axis. We take the Néel vector $\hat{\boldsymbol{n}}=\hat{\boldsymbol{y}}$ throughout this work unless otherwise specified. μ is the chemical potential and Δ_s the s-wave pairing amplitude.

For the Bogoliubov–de Gennes (BdG) Hamiltonian of Eq. (1), its momentum-space kernel reads

$$h_{\rm BdG}(\boldsymbol{k}) = \begin{pmatrix} h(\boldsymbol{k}) & i\Delta_s s_y \\ -i\Delta_s^* s_y & -h^T(-\boldsymbol{k})\end{pmatrix}, \tag{2}$$

under the basis $\Psi_{\boldsymbol{k}}=(C_{\boldsymbol{k}},C_{-\boldsymbol{k}}^\dagger)^T$, with the spinor $C_{\boldsymbol{k}}=(c_{A\boldsymbol{k}\uparrow},c_{A\boldsymbol{k}\downarrow},c_{B\boldsymbol{k}\uparrow},c_{B\boldsymbol{k}\downarrow})$, where $h(\boldsymbol{k})=h_{\rm KM}(\boldsymbol{k})+f_{\rm AM}(\boldsymbol{k},\chi)\sigma_0 s_y+M\sigma_z s_0-\mu$. Here, σ_z acts on the sublattice degrees of freedom. σ_0 and s_0 denote the two-dimensional identity matrices. The Kane–Mele Hamiltonian explicitly reads $h_{\rm KM}(\boldsymbol{k})=b_{x0}\sigma_x s_0+b_{y0}\sigma_y s_0+b_{zz}\sigma_z s_z$, where $b_{x0}=-t\big[\cos k_y'+2\cos k_x'\cos(k_y'/2)\big]$, $b_{y0}=t\big[\sin k_y'-2\cos k_x'\sin(k_y'/2)\big]$, $b_{zz}=2t'\big[\sin 2k_x'-2\sin k_x'\cos(3k_y'/2)\big]$ with $k_x'=k_x/2$ and $k_y'=k_y/\sqrt{3}$. The altermagnetism form factor in momentum space reads

$$f_{\rm AM}(\boldsymbol{k},\chi)=2\lambda\sum_{n=1}^{3}\cos\left[2(\theta_n-\chi)\right]\cos(\boldsymbol{k}\cdot\boldsymbol{a}_n), \tag{3}$$

where $\boldsymbol{a}_1=(1,0)$ and $\boldsymbol{a}_{2,3}=(1/2,\pm\sqrt{3}/2)$, with $\theta_n=\arg\boldsymbol{a}_n$. The choices $\chi=0$ and $\chi=\pi/4$ yield the $d_{x^2-y^2}$ and d_{xy} textures shown in Figs. 1(b) and 1(c), respectively.

The honeycomb flake has bearded edges along 0° and $\pm60^\circ$, with alternating A and B terminations, as illustrated in Fig. 2(b). The upper (lower) horizontal edge terminates on B (A). We also consider a bearded triangle with all three edges terminating on A. The lowest-energy states of this bearded hexagon exhibit three distinct localization patterns as the altermagnetic strength λ increases as shown in Fig. 2: extended along horizontal edges, localized at four corners, and paired at the two lateral vertices. The two corner-localized phases host four near-zero-energy levels, with protection mechanisms distinct from those of the four-corner localized phase.

III. LOW-ENERGY EDGE THEORY AND THE ORIGIN OF MAJORANA CORNER MODES

At the bulk valleys, the d-wave altermagnetic form factor in Eq. (3) cancels out. Edge projections depend on the boundary profile and sublattice termination [30, 49]. In the present model, this boundary dependence lifts the bulk cancellation and produces anisotropic edge masses. For bearded edges, the nearest-neighbor zero mode at $k_\parallel=0$ takes amplitudes $\phi_n=(\sqrt{3}/2)(-1/2)^n$ on the terminating sublattice. Here $n=0,1,\ldots$ labels rows moving inward from the boundary. We adopt this profile at leading order in t'/t. The row overlaps read $S_0=\sum_n\phi_n^2=1$ and $S_1=\sum_n\phi_n\phi_{n+1}=-1/2$. For an edge oriented at angle θ_j relative to $\boldsymbol{a}_1$, we define $\lambda_\parallel=\lambda\cos[2(\theta_j-\chi)]$ and $\lambda_\pm=\lambda\cos[2(\theta_j\pm\pi/3-\chi)]$. The identity $\lambda_++\lambda_-=-\lambda_\parallel$ then yields the projected contribution from the altermagnetic order

$$\begin{aligned} m_j &= 2\lambda_\parallel S_0+2(\lambda_++\lambda_-)S_1 \\ &= 3\lambda\cos[2(\theta_j-\chi)]. \end{aligned} \tag{4}$$

The negative adjacent-row overlap drives constructive addition between intra-row and inter-row contributions, yielding a finite edge gap when the sublattice potential u_j and superconductor pairing Δ_s are neglected.

Projection of the staggered and chemical potentials yields $u_j=\xi_j M-\mu$, where $\xi_j=+1$ (-1) labels A

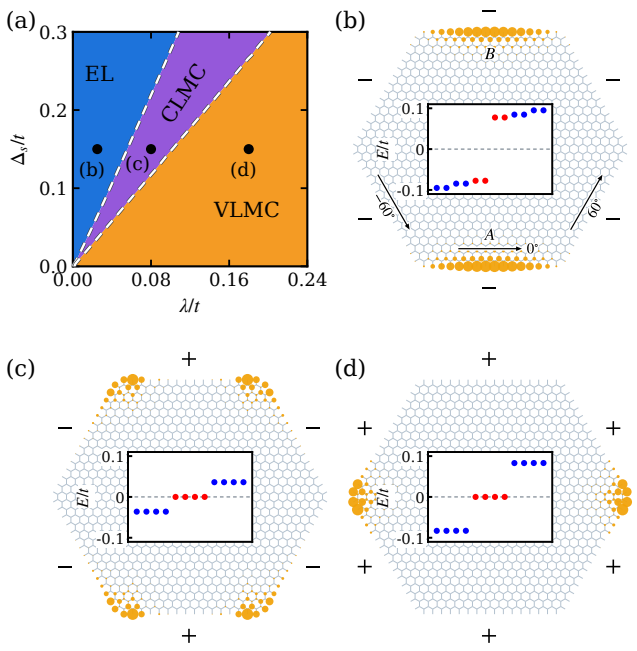


FIG. 2. (a) Phase diagram. EL denotes the edge-localized phase, CLMC denotes the corner-localized Majorana phase, and VLMC denotes the vertex-localized Majorana phase. The EL, CLMC, and VLMC phases are separated by the phase boundaries $|\Delta_s| = 3\lambda$ and $|\Delta_s| = 3\lambda/2$. (b) Density distributions of edge bound states (red dots in the inset) for a trivial superconductor. A and B denote A- and B-sublattice edge terminations. (c)–(d) Density distributions of Majorana corner modes in the CLMC and VLMC phases. The radii of the orange disks are proportional to the local density. The insets show the energy levels. In (b)–(d), the parameters $|\Delta_s|$ and λ are highlighted by black points in (a). In all panels, $M = \mu = \chi = 0$, and $t' = 0.03t$.

(B) termination. Onsite spin-singlet pairing maintains its amplitude because the unperturbed time-reversed edge-state partners carry unit orbital overlap. For the uniform superconducting order $\Delta_s = |\Delta_s|e^{i\phi}$ in Eq. (2), the Nambu rotation $U_\phi = e^{i\phi\tau_z/2}$ satisfies $U_\phi^\dagger h_{\rm BdG}(\boldsymbol{k},\phi)U_\phi = h_{\rm BdG}(\boldsymbol{k},0)$. Therefore without loss of generality, we adopt this real-pairing gauge for all single-flake results below, unless stated otherwise. Within the Nambu subspace of projected edge states, the edge Hamiltonian reads

$$h_{edge,j}(q) = v_0 q s_z + m_j s_y + u_j \tau_z - |\Delta_s|\tau_y s_y, \qquad (5)$$

where τ_α are Pauli matrices acting on Nambu space. Boundary lengths are measured counterclockwise in lattice units. $v_0 = -3t'$ denotes the Fermi velocity. $q = k - k_0$ is the small momentum, with $k_0 = 0$ marking the gapless point of the edge states when m_j, u_j, and Δ_s vanish. We retain terms linear in q, assuming a gapped bulk and an isolated edge doublet.

For the full edge Hamiltonian as in Eq. (5), its determinant takes the form

$$\det(h_{edge,j}) = (v_0^2 q^2 + m_j^2 - R_j^2)^2 + 4|\Delta_s|^2 v_0^2 q^2, \qquad (6)$$

where $R_j = \sqrt{u_j^2 + |\Delta_s|^2}$. For nonzero pairing, Eq. (6) vanishes only at $q = 0$ and $|m_j| = R_j$. At this gap-closing point $q = 0$, projection onto $s_y = \sigma = \pm 1$ yields a Nambu doublet $H_{j\sigma}(0) = \sigma m_j + u_j\tau_z - \sigma|\Delta_s|\tau_y$, with eigenenergies $E_{\sigma,\eta}(0) = \sigma m_j + \eta R_j$ $(\eta = \pm 1)$.

The competition between altermagnetism and superconducting pairing is captured by the effective edge mass

$$m_{{\rm edge},j} = |m_j| - \sqrt{u_j^2 + |\Delta_s|^2}, \qquad (7)$$

This quantity changes sign across the gap-closing transition. Its sign distinguishes the two distinct gapped edge phases (See Appendix A for details). The onsite potential u_j modifies the critical threshold from $|\Delta_s|$ to R_j, whereas the global superconducting phase has no effect. Its magnitude gives the minimum excitation energy at $q = 0$, but need not equal the global gap.

Near a gap-closing point, projection onto the two critical states yields a Dirac Hamiltonian. Matching neighboring edge bases at the corner and choosing positive velocity $v(\ell)$ gives its Hermitian real-space form,

$$h = -\frac{i}{2}\{v(\ell), \partial_\ell\}\widetilde{\varrho}_z + m_{\rm edge}(\ell)\widetilde{\varrho}_y, \qquad (8)$$

where $\widetilde{\varrho}_\alpha$ are Pauli matrices acting on the critical doublet, and the corner lies at $\ell = 0$. Resolving $\widetilde{\varrho}_x = \beta = \pm 1$, the zero-energy equation for the wave-function component ψ_β reads $\partial_\ell(\sqrt{v}\,\psi_\beta) = -\beta\frac{m_{\rm edge}}{v}(\sqrt{v}\,\psi_\beta)$. Wave-function decay on both sides requires $\beta m_{\rm edge}(0^+) > 0$ and $\beta m_{\rm edge}(0^-) < 0$. The mass domain wall therefore hosts a Jackiw–Rebbi zero mode [25, 50]. Particle–hole symmetry allows its full Nambu spinor to be chosen self-conjugate, identifying it as a Majorana mode.

For long gapped edges, we define $r_j = \mathrm{sgn}\big(m_{{\rm edge},j}\big)$. The protected Majorana count is $N_{\rm prot} = \frac{1}{2}\sum_{j=1}^{6}|r_{j+1} - r_j|$, with $r_7 = r_1$. This formula counts protected single Majorana modes at each edge-phase interface. Additional pairs can appear when an extra symmetry suppresses their hybridization. The detailed projection and zero-mode envelope are given in Appendix A. Overlap between well-separated corner modes induces exponentially small finite-energy splittings.

The orientation dependence of m_j yields distinct gap-closing conditions for horizontal and oblique ribbons as in Fig. 3. For $M = \mu = \chi = 0$, the edge masses read $m_{{\rm edge},h} = 3|\lambda| - |\Delta_s|$ and $m_{{\rm edge},o} = 3|\lambda|/2 - |\Delta_s|$. When $|\Delta_s| > 3|\lambda|$, both masses are negative. No mass domain wall emerges in this regime, and the lowest-energy states carry finite energy with spectral weight concentrated on horizontal edges, where the edge gap is smaller as in Fig. 2(b). Upon increasing $|\lambda|$, the horizontal edge mass changes sign first, while the oblique mass remains negative. This defines the four-corner phase window

$$\frac{1}{3}|\Delta_s| < |\lambda| < \frac{2}{3}|\Delta_s|. \qquad (9)$$

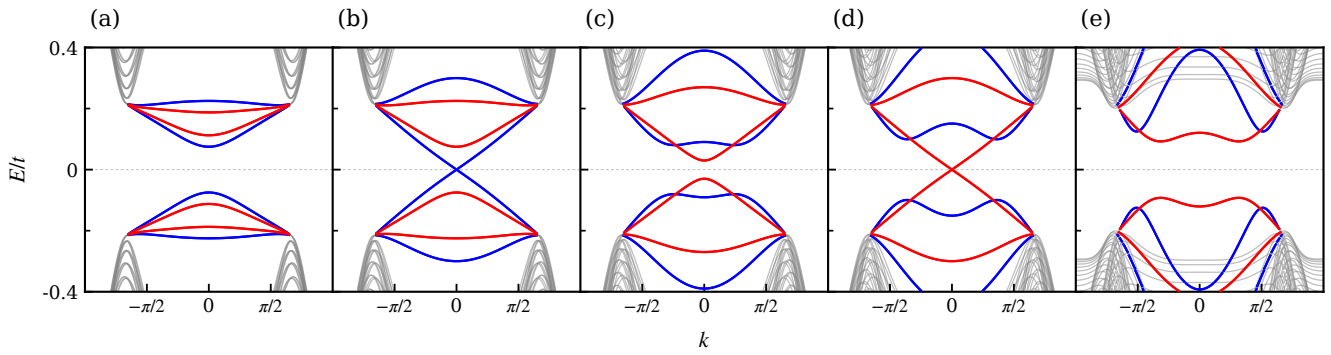


FIG. 3. Bearded-ribbon quasiparticle spectra at $M = \mu = \chi = 0$ and $|\Delta_s| = 0.15t$. (a)–(e) $\lambda/t = 0.025, 0.050, 0.080, 0.100, 0.180$, respectively. Blue and red curves correspond to horizontal (0°) and oblique (±60°) edge bands. Gray curves represent the remaining bulk bands. The spectrum for the −60° edge follows from the 60° result under $k \mapsto -k$. The horizontal-edge gap closes in panel (b), and the oblique-edge gap closes in panel (d).

A Majorana mode localizes at each of the four horizontal endpoints within this window [Fig. 2(c)]. At $|\Delta_s| = 0.15t$, our low-energy theory predicts gap closings at $\lambda = 0.05t$ and $0.10t$, consistent with the numerical results in Figs. 3(b) and 3(d). These two thresholds delineate the four-corner region in the phase diagram Fig. 2(a).

For $|\Delta_s| < 3|\lambda|/2$, both edge masses become positive and $N_{\rm prot} = 0$. Nevertheless, four near-zero-energy states persist in this phase, as seen in Fig. 2(d). At $M = \mu = 0$, the lattice Hamiltonian commutes with $Q = \tau_y\sigma_z$ in the real-pairing gauge. σ_z acts on the sublattice degrees of freedom. In the edge theory, each lateral vertex hosts a pair of Majorana modes in different Q sectors, whose mixing is forbidden by Q conservation (see Appendix A). The term $-\mu\tau_z$ couples the two sectors and lifts the degeneracy of each vertex mode pair without closing the edge gap. These vertex-bound mode pairs are therefore qualitatively distinct from the four isolated domain-wall Majorana corner modes shown in Fig. 2(c).

Changing the edge termination modifies the boundary thresholds. For zigzag edges, an analogous mass criterion as in Eq. (7) holds, with $m_j^{(z)} = -2p\lambda\cos[2(\theta_j - \chi)]$ and $u_j^{(z)} = \xi_j pM - \mu$, where $p = t/\sqrt{t^2 + 16t'^2}$. The pairing amplitude remains unaltered. These coefficients encode the finite spin-orbit-coupling dependence inherited from the unperturbed edge states. At $M = \mu = \chi = 0$, the four-corner phase window reads $|\Delta_s|/(2p) < |\lambda| < |\Delta_s|/p$. The derivation of this projected boundary condition is provided in Appendix A.

For bearded-edge flakes, opposite edges satisfy $m_{j+3} = m_j$ while terminating on opposite sublattices. Their transition thresholds obey $R_{\rm B}^2 - R_{\rm A}^2 = 4M\mu$, where $R_{\rm A,B}$ correspond to R_j for A-type and B-type edge terminations. Nonzero M and μ applied simultaneously can split the transition points on opposite edges. This enables selective tuning of corner modes (See Appendix B), which underpins the braiding protocols of Majorana corner modes.

IV. TUNABLE MAJORANA CORNER MODES AND CLIFFORD QUANTUM GATES

Sign reversal of an edge mass transfers a Majorana mode between its endpoints, provided the two adjacent edges carry opposite mass signs. The onsite potentials set the threshold for this reversal, while the altermagnetic texture determines the exchange projection on each edge. Together, these controls select which edge undergoes a transition and thereby which pair of corners is connected by the transfer. The boundary criterion in Eq. (7) thus provides a control principle for routing Majorana corner modes.

We first examine the transfer and exchange of an isolated Majorana pair on a single triangular flake, then extend the analysis to interconnected flakes that provide the additional modes and transfer branches required for encoded Clifford gates.

For such connected device networks, we adopt

$$H_{\rm net}(s) = \sum_a H_a(s) - t_c \sum_{(ai,bj)\in\mathcal{L}} \left(c_{ai}^\dagger c_{bj} + \text{H.c.} \right), \quad (10)$$

where H_a denotes the island Hamiltonian given in Eq. (1), c_{ai} is the electron spinor, and $\mathcal{L}$ contains each spin-conserving inter-island bond exactly once. The tunnel amplitude is real and fixed at $t_c = 0.30t$. The parameter $s \in [0,1]$ parametrizes the control trajectory $\{\chi_a(s), M_a(s), \mu_a(s)\}_a$, with physical time Ts. Each island supports uniform s-wave superconducting pairing $\Delta_{s,a} = |\Delta_s| e^{i\phi_a}$ with fixed ϕ_a. Adopting the real-pairing gauge independently on each island yields

$$\widetilde{T}_{ab} = -t_c \tau_z e^{i(\phi_b - \phi_a)\tau_z/2}, \quad (11)$$

with the spin identity matrix omitted. Fixed pairing-phase differences enter the inter-island tunnel coupling, whereas edge masses govern the mode-transfer pathways on each individual flake. Taken together, the low-energy edge theory and the full network BdG Hamiltonian de-

scribe the propagation of corner modes across these interconnected devices, as detailed in the subsequent subsections.

A. Braiding of two Majorana corner modes

The triangular geometry has three edges, each associated with a distinct altermagnetic angle θ_j. Successive sign reversals of edge mass route two modes through the unoccupied vertex [31]. For an A-terminated bearded triangular flake, all edges share the same threshold $R = \sqrt{(M-\mu)^2 + |\Delta_s|^2}$. The three altermagnetic induced edge masses read $3|\lambda\cos[2(\theta_j - \chi)]|$ with $\theta_{j=1,2,3} = 0, \pi/3, 2\pi/3$. The smallest never exceeds $3|\lambda|/2$, and the largest is at least $3\sqrt{3}|\lambda|/2$. The inequality

$$\frac{3}{2}|\lambda| < \sqrt{(M-\mu)^2 + |\Delta_s|^2} < \frac{3\sqrt{3}}{2}|\lambda| \tag{12}$$

maintains mixed edge-mass signs for all noncritical orientations. This condition produces two mass domain walls. At each reversal event, one mode spreads along the reversing edge, and then relocates to the opposite endpoint.

For the parameters in Fig. 4, the bottom-edge mass starts positive, while the two oblique-edge masses start negative. Upon increasing χ, the right-edge mass reverses first, followed by the bottom-edge mass and then the left-edge mass. The first reversal shifts γ_2 from the lower-right corner to the top vertex, with γ_1 remaining localized at the lower-left corner. The second reversal transports γ_1 to the vacant lower-right corner. The third reversal moves γ_2 to the lower-left corner. The two Majorana modes swap their positions at $\chi = \pi/4$. This sequence is illustrated in Figs. 4(a)–(d). Along this control path, the spectrum of the finite triangular flake hosts an isolated near-zero-energy Majorana pair as shown in Fig. 4(e). Edge delocalization during each mass-sign reversal establishes the spatial connection between the initial and final corners.

At $\chi = \pi/4$, the two modes occupy one another's initial corners, and all three edge masses undergo sign reversal. Proceeding to $\chi = \pi$ completes the texture period and recovers the Hamiltonian for a fixed Néel vector. Parallel transport across this full cycle yields

$$\gamma_1 \mapsto -\gamma_1, \quad \gamma_2 \mapsto -\gamma_2. \tag{13}$$

See Appendix C for more details. This overall sign change arises from the geometric action associated with the full texture cycle. The four spatial interchanges partition the parameter path into successive gapped segments. Introducing a second Majorana pair enlarges the fixed-parity sector to form a two-dimensional logical subspace. Braiding within this subspace generates qubit rotations. Transport of paired Majorana corner modes in a bearded hexagonal flake is also demonstrated in Appendix B.

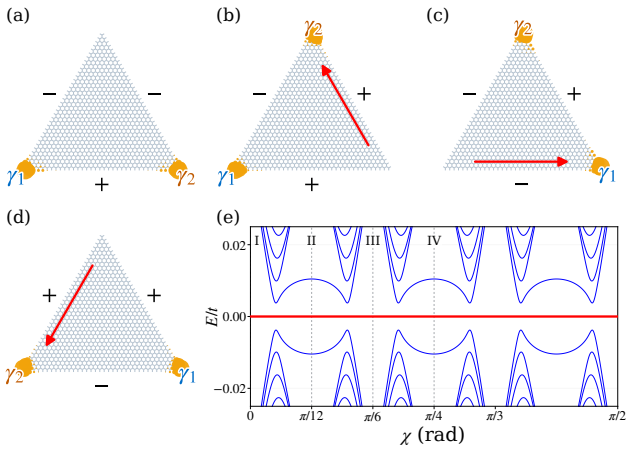


FIG. 4. Braiding of two Majorana corner modes in an A-terminated bearded triangle flake with 1404 lattice sites. (a)–(d) Density distributions of Majorana corner modes at $\chi = 0, \pi/12, \pi/6, \pi/4$, respectively. Red arrows indicate the transfer directions of these Majorana corner modes. The $+$ and $-$ signs denote positive and negative effective edge masses $m_{\mathrm{edge},j}$, respectively. (e) Low-energy BdG spectrum for $0 \le \chi \le \pi/2$. Red curves mark the two Majorana zero-energy levels. Guidelines I–IV mark the χ values used in panels (a)–(d). The common parameters are $\lambda = 0.09t$, $|\Delta_s| = 0.15t$, $\mu = 0.10t$, $M = 0$, and $t' = 0.03t$.

B. X gate

In this subsection, we present an example demonstrating that Majorana braiding can also be implemented on hexagonal flakes to realize the X gate. The two coupled hexagonal flakes in Fig. 5(a) host four Majorana corner modes, initially localized at the horizontal-edge endpoints of island L, with island R serving as an auxiliary transfer branch. We define the complex fermions

$$f_1 = \frac{\gamma_1 + i\gamma_2}{2}, \qquad f_2 = \frac{\gamma_3 + i\gamma_4}{2}, \tag{14}$$

with occupation-number operators $n_j = f_j^\dagger f_j$. The logical qubit is encoded in the even-parity sector as

$$|0\rangle_{\mathrm{L}} = |00\rangle, \qquad |1\rangle_{\mathrm{L}} = |11\rangle, \tag{15}$$

where $|11\rangle = f_1^\dagger f_2^\dagger |00\rangle$, and L denotes the logical basis. We adopt $U_{ij} = \exp(\pi\gamma_i\gamma_j/4)$ for an oriented braiding operator and $U_{ij}^\dagger$ for its inverse [51]. A Majorana mode γ_k transforms under the braiding operator U_{ij} as $U_{ij}\gamma_k U_{ij}^\dagger$. All braiding matrices are defined relative to a common initial Majorana frame prior to composition, where the rightmost factor acts first. Generator indices mark fixed reference positions, and colors in the diagrams track the propagating modes along the control path. These generators specify the ideal logical transformation. Parallel transport quantifies its physical implementation along the device control trajectories.

Each hexagonal flake features zigzag termination along its two right oblique edges and bearded termination along the remaining edges. At the fixed texture angle $\chi = \pi/4$, the projection of the altermagnetic form factor along horizontal edges vanishes, and the horizontal edge masses remain negative. Mixed edge terminations yield distinct thresholds for edge-mass sign reversal on the oblique edges. Using Eq. (7) and the parameter p introduced in Sec. III, the thresholds for edge-mass sign reversal occur at $|\xi_j p M_a - \mu_a| = 0.203t$ for zigzag edges and $|\xi_j M_a - \mu_a| = 0.360t$ for bearded edges. Tuning M_a and μ_a crosses the thresholds of the right oblique edges while the left oblique masses stay positive. This selects γ_2 and γ_3 for transport and pins γ_1 and γ_4 at their original corners.

The auxiliary island permits one mode to be diverted from the main braiding path while the other proceeds onward. For $0 \le s \le 1/2$, we first flip the sign of the lower-right edge mass of island L, which brings γ_3 to the junction. A pulse applied to M_R, with $M_R + \mu_R$ kept constant, transfers γ_3 to the upper-left corner of island R, as shown in Figs. 5(b) and (c). We then flip the sign of the upper-right edge mass of island L to bring γ_2 toward the junction. Restoring the original sign of the lower-right edge mass moves γ_2 into the initial corner of γ_3, as depicted in Figs. 5(d) and (e). Removing the pulse transports γ_3 back through the junction to the upper-right corner of island L, as illustrated in Figs. 5(f) and (g). All control parameters return to their starting values at $s = 1/2$, completing one braiding operation.

For this chosen orientation, the elementary braiding yields

$$U_{23}\gamma_2 U_{23}^\dagger = -\gamma_3, \qquad U_{23}\gamma_3 U_{23}^\dagger = \gamma_2, \tag{16}$$

while γ_1 and γ_4 remain unchanged. Restricted to the even-parity subspace defined by Eq. (15), $U_{23}|_{\rm even} = (I + iX)/\sqrt{2}$. Repeating this braiding operation with identical orientation returns the Majoranas to their original corners while flipping the sign of γ_2 and γ_3. Consequently, $f_1 \mapsto f_1^\dagger$ and $f_2 \mapsto -f_2^\dagger$: the occupation of both complex fermions flips, while the total fermion parity remains unchanged. The resulting logical action reads

$$U_X|_{\rm even} = U_{23}^2|_{\rm even} = iX. \tag{17}$$

Two successive braiding operations map this geometrically induced sign transformation into a logical basis flip [52]. The two stationary Majoranas provide the parity partners required for this gate action.

C. Hadamard gate

Access to a second logical rotation axis is gained by exchanging the two Majorana modes of an encoded fermion, complementing the rotation supplied by the X-gate protocol. This operation yields $U_{12}|_{\rm even} = e^{i\pi Z/4}$, with $Z|n\rangle_{\rm L} = (-1)^n|n\rangle_{\rm L}$. The three-triangle flake network

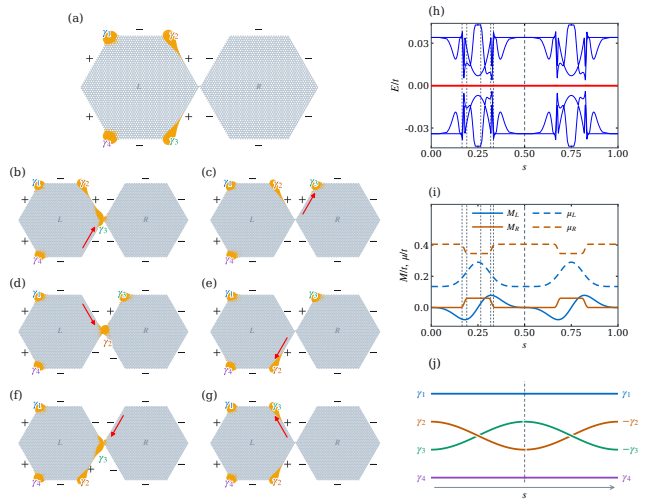


FIG. 5. X gate via braiding in two coupled mixed-termination hexagonal flakes. Each flake contains 3850 lattice sites. The two flakes are connected by a single bond at their adjacent tips. (a)–(g) Density distributions of Majorana modes during one braiding exchange, $0 \le s \le 1/2$. Labels track the Majorana modes. Arrows indicate transfer directions. The + and − signs denote positive and negative effective edge masses $m_{{\rm edge},j}$, respectively. (h) Low-energy BdG energies along the full two-exchange braiding path. Red curves mark Majorana zero mode levels and blue curves mark the remaining energy levels. Dotted guidelines locate the snapshots (a)–(g). (i) On-site controls along the same path. Solid and dashed curves denote controls M_a and μ_a. (j) Braiding sequence along the full two-exchange path. Endpoint signs are defined with respect to the initial reference frame. Each half repeats the same closed control loop. The parameters are $t' = 0.06t$, $\lambda = |\Delta_s| = 0.15t$, $\chi_L = \chi_R = \pi/4$, $t_c = 0.30t$ and pairing phases $(\phi_L, \phi_R) = (0, \pi/3)$. Initially, $M_L = M_R = 0$, $\mu_L \simeq 0.134t$, and $\mu_R \simeq 0.405t$.

in Fig. 6(a) makes both U_{12} and U_{23} accessible in the encoding of Eq. (15). Central island A provides a transfer pathway between islands B and C. The Majorana zero mode at the lower tip of B remains a spectator.

The control sequence combines texture rotations, which modify the altermagnetic projection, with chemical-potential pulses, which tune the competing threshold $\sqrt{\mu_a^2 + |\Delta_s|^2}$ at $M_a = 0$. Reducing μ_A supports domain-wall formation on the central island. The first and third stages tune χ_A and χ_B to activate the B–A branch. The intermediate stage fixes χ_B and applies a pulse on μ_C to open the transport route across the A–C junction, as depicted in Fig. 6(c). All control parameters return to their initial values after the three-stage sequence. These three stages implement $U_{12}^\dagger$, $U_{23}^\dagger$, and $U_{12}^\dagger$ in chronological order, as depicted in Fig. 6(d). These noncommuting logical rotations yield $U_H = U_{12}^\dagger U_{23}^\dagger U_{12}^\dagger$ with

$$U_H|_{\rm even} = -\frac{i}{\sqrt{2}}(X + Z) = -iH, \tag{18}$$

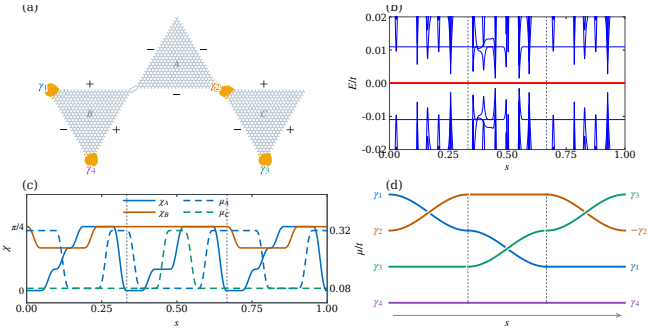


FIG. 6. Hadamard gate via braiding in three coupled bearded triangle flakes. (a) Initial density distribution on the flakes with labels. The signs + and − indicate positive and negative effective edge masses $m_{\mathrm{edge},j}$ in the initial configuration. Each flake contains 648 lattice sites. Each junction contains two interflake bonds with hopping strength t_c. (b) Low-energy BdG energies along the closed transport path. Energy curves follow the color convention of Fig. 5(h). (c) Control parameters along the same path. Solid curves correspond to χ_A, χ_B. Dashed curves correspond to μ_A, μ_C on the right vertical scale. (d) Braiding sequence. Colors track the moving Majorana modes. Rows mark the initial reference positions. Endpoint signs are defined with respect to this reference frame. The parameters are $t' = 0.03t$, $\lambda = 0.09t$, $|\Delta_s| = 0.15t$, $M_a = 0$, $t_c = 0.30t$ and pairing phases $(\phi_A, \phi_B, \phi_C) = (0, 2\pi/3, 2\pi/3)$. The fixed control parameters are $\chi_C = \pi/4$ and $\mu_B = 0.08t$. Initial values of all control parameters are listed in Appendix C.

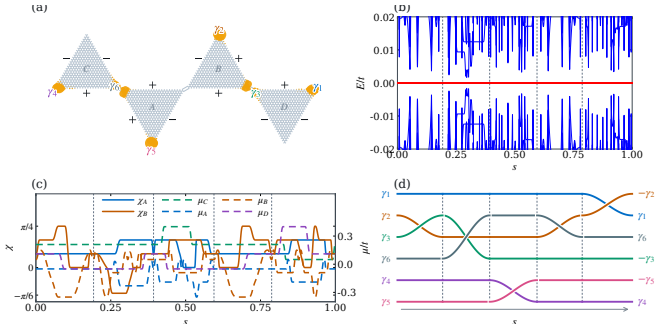


FIG. 7. CNOT gate via braiding in four coupled bearded triangle flakes. These flakes share the same lattice size, t', λ, $|\Delta_s|$, and interflake hopping bonds as in Fig. 6. (a) Initial density distributions of the six Majorana modes in the C–A–B–D triangle flakes. + and − denote the signs of the effective edge masses $m_{\mathrm{edge},j}$ in the initial configuration. (b) Low-energy BdG energies. Energy curves adopt the same color scheme as in Fig. 5(h). (c) Control parameters. Solid curves represent χ_A, χ_B. Dashed curves correspond to the four chemical potentials on the right vertical scale. (d) Braiding sequence with five exchanges. This panel adopts the same conventions as in Fig. 6(d). The row order is $1, 2, 3, 6, 4, 5$. The parameters are $M_a = 0.05t$ and pairing phases $(\phi_C, \phi_A, \phi_B, \phi_D) = (0, 2\pi/3, 0, 2\pi/3)$. The fixed anisotropy textures are $\chi_C = \chi_D = \pi/30$. Initial values of all control parameters are listed in Appendix C.

where $H = (X+Z)/\sqrt{2}$. Dropping the global U(1) phase factor, the logical-state transformation reads $|0\rangle_{\mathrm{L}} \mapsto \frac{1}{\sqrt{2}}\big(|0\rangle_{\mathrm{L}} + |1\rangle_{\mathrm{L}}\big)$, $|1\rangle_{\mathrm{L}} \mapsto \frac{1}{\sqrt{2}}\big(|0\rangle_{\mathrm{L}} - |1\rangle_{\mathrm{L}}\big)$. Successive rotations about the two distinct logical axes generate equal-amplitude superpositions, carrying the input-dependent relative sign characteristic of the Hadamard gate.

D. CNOT gate

To construct a conditional gate, we employ six Majorana zero modes within a common even-parity sector [10]. The four coupled triangle flakes C–A–B–D in Fig. 7(a) implement the encoding

$$\begin{aligned} f_j &= \frac{\gamma_{2j-1} + i\gamma_{2j}}{2}, \qquad j = 1, 2, 3, \\ |a, b\rangle_{\mathrm{L}} &= |a, a \oplus b, b\rangle, \qquad a, b \in \{0, 1\}, \end{aligned} \tag{19}$$

where the first line defines the three encoded fermion annihilation operators, and the second line gives the logical basis. The fermion creation operators follow the ordering $f_1^\dagger, f_2^\dagger, f_3^\dagger$. Here a labels the control bit and b the target bit. $\oplus$ denotes addition modulo two. This is a shared-parity encoding. The middle occupation $n_2 = a \oplus b$ correlates the two logical bits, while $n_1 = a$ and $n_3 = b$ remain independent.

The central A–B flake link connects the two transfer branches C–A and B–D. Chemical-potential pulses on the outer islands and rotations of χ_A, χ_B drive domain walls along these branches. This sequence arranges the five braiding operations $U_{23}^\dagger$, $U_{26}^\dagger$, $U_{45}^\dagger$, U_{23}, and $U_{12}^\dagger$ in chronological order. After the first braiding, reference site 2 hosts the mode originally labeled γ_3. As a result, $U_{26}^\dagger$ corresponds to the crossing of the trajectories of γ_3 and γ_6, as depicted in Fig. 7(d). All control parameters return to their initial values after the fifth stage. These ordered braiding operations yield

$$U_{\mathrm{CNOT}} = U_{12}^\dagger U_{23} U_{45}^\dagger U_{26}^\dagger U_{23}^\dagger \tag{20}$$

within the common reference frame. The induced fermion transformation reads

$$f_1 \mapsto -i f_1, \qquad f_2 \mapsto i f_3, \qquad f_3 \mapsto i f_2, \tag{21}$$

where these maps represent $U_{\mathrm{CNOT}} f_j U_{\mathrm{CNOT}}^\dagger$. This transformation conserves n_1 and swaps n_2 and n_3. Because the occupations n_2 and n_3 differ if and only if $a = 1$, their interchange flips the target bit b precisely when the control bit is set. Accounting for phase contributions from fermionic reordering yields

$$U_{\mathrm{CNOT}} |a, b\rangle_{\mathrm{L}} = e^{-i\pi/4} |a, a \oplus b\rangle_{\mathrm{L}}. \tag{22}$$

All four logical input states acquire the same global phase. This produces the controlled bit flip while preserving the relative logical phases. Up to this common phase, the gate maps $\frac{1}{\sqrt{2}}\big(|00\rangle_{\mathrm{L}} + |10\rangle_{\mathrm{L}}\big)$ to $\frac{1}{\sqrt{2}}\big(|00\rangle_{\mathrm{L}} + |11\rangle_{\mathrm{L}}\big)$. This transformation generates entanglement between the two encoded logical qubits.

The gapped energy spectra in Figs. 5–7 preserve the selected four- or six-mode subspaces under the sampled control parameters. Parallel transport with matched endpoints yields logical unitary matrices that closely approximate the target X, H, and CNOT gates, including their relative phases. The phase-minimized Frobenius distances read $\delta_X = 5.65 \times 10^{-3}$, $\delta_H = 2.04 \times 10^{-5}$, and $\delta_{\mathrm{CNOT}} = 1.87 \times 10^{-4}$. Appendix C details how the CNOT gate arises from composing the five braided transport matrices within the common reference frame. This appendix also provides the energy scales and driving conditions required for coherent gate operation. These control trajectories combine sign reversals of the edge mass to realize encoded single-qubit rotations and the conditional two-qubit gate.

V. DISCUSSION AND CONCLUSION

Directional altermagnetic order yields a finite edge projection despite canceling contributions at the bulk valleys. On the honeycomb lattice, edge orientation and termination jointly determine this projection and the resulting edge masses. Bearded and zigzag edges obey a similar edge-mass criterion but differ in their threshold values, which gives rise to Majorana corner modes. Opposite edges of a hexagonal flake undergo simultaneous sign reversal of the edge mass when their thresholds coincide, while triangular flakes support successive mode transfers through a third vertex. During transfer, the Majorana mode spreads along the edge whose mass reverses sign. By organizing edge-mass reversals into closed control cycles, we translate spatial Majorana transport into geometric operations within fixed-parity encodings, realizing the X, Hadamard, and CNOT gates with the required relative logical phases.

The model platform is a Kane–Mele layer proximitized by an s-wave superconductor and a d-wave altermagnet. The superconducting response of altermagnet–superconductor hybrids has been studied through their thermodynamic properties [53] and finite-momentum Cooper pairing [54]. Separate material building blocks for this heterostructure have experimental precedents. Bismuthene provides a promising honeycomb quantum spin Hall candidate [55], while monolayer WTe_2 shows signatures of proximity-induced pairing along its helical edge [56]. These systems separately validate the key ingredients of quantum spin Hall transport and superconducting proximity effect required for our heterostructure. Reported altermagnetic band splitting [57–59] and d-wave magnetic order in KV_2Se_2O [60] suggest a viable material basis for directional altermagnetism, while domain manipulation in MnTe [61] illustrates the possibility of tuning magnetic configurations.

Possible control mechanisms include superconducting phase bias, strain, and electrostatic gating. Superconducting phase differences between neighboring islands can be set by phase bias, as demonstrated in phase-controlled Josephson junctions [62]. Strain engineering provides a possible route to tuning the altermagnetic anisotropy, as supported by theoretical studies of strained altermagnet–superconductor heterostructures [63]. Local electrostatic gates can tune the chemical potential μ [64], while a perpendicular electric field can generate a staggered potential M in buckled honeycomb layers with sublattice-dependent electric-field coupling [65]. A functional device would require integration of these components into compatible heterointerfaces with suitable edge terminations while preserving proximity-induced pairing.

Taken together, our edge-mass control scheme provides a pathway for the manipulation of Majorana corner modes and the realization of Clifford gates in altermagnet–superconductor heterostructures.

VI. ACKNOWLEDGMENT

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Appendix A: Edge projection and edge mass domain walls

1. Edge projection

We adopt the real-pairing gauge used in the main text. The projection keeps terms linear in momentum and to leading order in λ, M, μ, and $|\Delta_s|$ within an isolated edge doublet. For bearded edges, we use the leading-order profile in t'/t. For zigzag edges, we retain the finite-t' dependence of the unperturbed edge states.

For a bearded edge parallel to $\boldsymbol{a}_1$, the nearest-neighbor recursion $t\phi_n + 2t\cos(k/2)\phi_{n+1} = 0$ gives $\rho_b(k) = -[2\cos(k/2)]^{-1}$, where $n \geq 0$ indexes rows inward from the boundary. The normalized wavefunction profile at the crossing $k = 0$ reads

$$|\phi_{j\sigma}\rangle = \frac{\sqrt{3}}{2} \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n |n, \xi_j, \sigma\rangle. \tag{A1}$$

Here $\sigma = \pm 1$ labels the s_z eigenstates and $\xi_j = \pm 1$ the terminating sublattice, on which the mode resides. The same-row overlap equals unity and the adjacent-row overlap equals ρ_b. We define $\lambda_{\parallel} = \lambda\cos[2(\theta_j - \chi)]$ and $\lambda_{\pm} = \lambda\cos[2(\theta_j \pm \pi/3 - \chi)]$, which satisfy $\lambda_+ + \lambda_- = -\lambda_{\parallel}$. The projected altermagnetic and intrinsic spin-orbit contributions read

$$\begin{aligned} h_{\mathrm{edge,AM}} &= 2\lambda_{\parallel}[\cos k - \rho_b \cos(k/2)]s_y, \\ h_{\mathrm{edge,SO}} &= -2t'[\sin k - 2\rho_b \sin(k/2)]s_z. \end{aligned} \tag{A2}$$

The directed-bond convention for the A-terminated reference edge is $(\nu_\parallel, \nu_+, \nu_-) = (1, -1, 1)$. At $k = 0$, $\rho_b = -1/2$, giving $m_j = 3\lambda_\parallel$ and the edge velocity $v_0 = -3t'$. Transforming the bond signs and momentum together gives the same velocity for all bearded edges when momentum is measured along the counterclockwise boundary direction. The onsite potential projects to $u_j = \xi_j M - \mu$. The unperturbed time-reversed edge-state partners have unit orbital overlap, so onsite pairing retains its amplitude $|\Delta_s|$. These results give Eq. (5) in the projected Nambu basis $(d_{jq\uparrow}, d_{jq\downarrow}, d^\dagger_{j,-q,\uparrow}, d^\dagger_{j,-q,\downarrow})^T$.

For the zigzag edge, the band crossing occurs at $k = \pi$, where finite spin-orbit coupling induces sublattice admixture. Let $a = 4t'/[\sqrt{t^2 + 16t'^2} + t]$. On an A-terminated reference edge, the orbital amplitudes are

$$\begin{aligned} \phi_{A,2n,\sigma} &= \sqrt{1 - a^2}(-a^2)^n, \\ \phi_{B,2n,\sigma} &= \sigma a\, \phi_{A,2n,\sigma}, \end{aligned} \tag{A3}$$

with zero amplitudes on odd rows. The sublattice polarization and the opposite-spin orbital overlap at the same momentum both equal $p = (1 - a^2)/(1 + a^2) = t/\sqrt{t^2 + 16t'^2}$. The adjacent-row altermagnetic matrix element vanishes because odd rows carry no weight. The same-row term therefore gives $-2p\lambda_\parallel$. For the reference edge, the nearest-neighbor and intrinsic spin-orbit terms contribute $4pt'$ and $2pt'$ to the velocity, respectively. The projected coefficients are

$$\begin{aligned} m_j^{(z)} &= -2p\lambda \cos[2(\theta_j - \chi)], \\ u_j^{(z)} &= \xi_j pM - \mu, \qquad |v_j^{(z)}| = 6pt'. \end{aligned} \tag{A4}$$

Onsite pairing links time-reversed edge states at $k = \pi$ and $k = -\pi$, whose orbital overlap remains unity. Thus, p renormalizes the altermagnetic contribution and staggered potential, but not $|\Delta_s|$.

For bearded edges, with $R_j^2 = u_j^2 + |\Delta_s|^2$, squaring Eq. (5) gives

$$E_\pm^2 = v_0^2 q^2 + m_j^2 + R_j^2 \pm 2\sqrt{m_j^2 R_j^2 + u_j^2 v_0^2 q^2}. \tag{A5}$$

The product $E_+^2 E_-^2$ recovers Eq. (6) and the gap-closing condition in the main text. The magnitude $|m_{\mathrm{edge},j}|$ gives the lowest excitation energy at $q = 0$, whereas the minimum over momentum can occur at finite q for $u_j \neq 0$. Within the projected Nambu basis, $B_j = h_{\mathrm{edge},j}(0)\tau_x$ is antisymmetric, with Pfaffian

$$\mathrm{Pf}\, B_j = m_j^2 - R_j^2 = (|m_j| + R_j) m_{\mathrm{edge},j}. \tag{A6}$$

With a common gapped high-momentum reference whose Pfaffian sign is unchanged, opposite edge-mass signs distinguish the two gapped edge phases [66, 67]. The same criterion applies to zigzag edges with m_j and u_j replaced by $m_j^{(z)}$ and $u_j^{(z)}$.

2. Mass domain walls and vertex modes

Near a gap-closing point, the Nambu overlap between the two critical states has magnitude $|\Delta_s|/R_j$. Projection onto this critical doublet gives

$$h_{\mathrm{crit},j}(q) = v_{\mathrm{crit},j} q \widetilde{\varrho}_z + m_{\mathrm{edge},j} \widetilde{\varrho}_y, \tag{A7}$$

with $|v_{\mathrm{crit},j}| = |v_0 \Delta_s|/R_j$. In the matched continuum Hamiltonian of Eq. (8), $v(\ell) > 0$ approaches $|v_{\mathrm{crit},j}|$ on each edge segment. Integrating the zero-energy equation gives

$$\psi_\beta(\ell) \propto \frac{|\widetilde{\varrho}_x = \beta\rangle}{\sqrt{v(\ell)}} \exp\left[-\beta \int_0^\ell \frac{m_{\mathrm{edge}}(\ell')}{v(\ell')}\, \mathrm{d}\ell'\right], \tag{A8}$$

where $\beta = \pm 1$. Normalizability requires $\beta m_{\mathrm{edge}} > 0$ on the $\ell > 0$ edge and $\beta m_{\mathrm{edge}} < 0$ on the $\ell < 0$ edge, with masses evaluated away from the corner. The resulting domain-wall state can be chosen self-conjugate under particle–hole symmetry, $\mathcal{C}\psi = \psi$. Its decay length near the gap closing is $|v_{\mathrm{crit},j}/m_{\mathrm{edge},j}|$; away from this limit, the decay exponents follow from the full edge Hamiltonian.

For $M = \mu = 0$, the lattice Hamiltonian commutes with $Q = \tau_y \sigma_z$ and anticommutes with $\mathcal{S} = \tau_y s_x$. Particle–hole symmetry preserves the global Q-sector label $\zeta = \pm 1$. On a bearded edge, $\tau_y = \xi_j \zeta$. With $\varrho_x = \xi_j s_x$, $\varrho_y = s_y$, and $\varrho_z = \xi_j s_z$, the sector Hamiltonian reads

$$h_{\mathrm{edge},j\zeta}(q) = V_j q \varrho_z + \mathcal{M}_{j\zeta} \varrho_y, \tag{A9}$$

where $V_j = \xi_j v_0$ and $\mathcal{M}_{j\zeta} = m_j - \xi_j \zeta |\Delta_s|$. We assume corner matching that preserves Q, $\mathcal{S}$, and probability current, with no additional corner orbitals. The zero-energy equation on each edge is then $\partial_\ell \psi = -\kappa_{j\zeta} \varrho_x \psi$, where $\kappa_{j\zeta} = \mathcal{M}_{j\zeta}/V_j = (\xi_j m_j - \zeta|\Delta_s|)/v_0$. Under this matching, each sector hosts a zero mode when $\kappa_{j\zeta}\kappa_{j+1,\zeta} < 0$. The sign of V_j enters the decay condition. The solutions on adjacent edges are matched piecewise, without interpolating V_j through zero.

Numbering the edge segments counterclockwise from the upper B-terminated horizontal edge gives $\xi_j = (-1)^j$. At $\chi = 0$, the quantities entering the sector decay condition are

$$(\xi_1 m_1, \ldots, \xi_6 m_6) = \frac{3}{2}\lambda(-2, -1, 1, 2, 1, -1). \tag{A10}$$

In the window of Eq. (9), one sector wall occurs at each horizontal endpoint, giving four isolated Majorana corner modes. For $0 < |\Delta_s| < 3|\lambda|/2$, both sectors have walls at junctions 2–3 and 5–6, giving two zero modes at each lateral vertex. These modes have opposite eigenvalues of $\mathcal{S} = \zeta \varrho_x$ and belong to different Q sectors. Conservation of Q forbids their mixing. A weak $-\mu\tau_z$ perturbation couples the sectors and can split the pair while preserving $\mathcal{S}$ and the edge gap. These additional pairs are not counted by N_{prot}.

The continuum description requires decay lengths much larger than the lattice spacing; well-separated corner modes further require much longer edges. For short decay lengths, quantitative splittings require lattice eigenstates. Finite-coupling corrections can shift the leading-order transition thresholds.

Appendix B: Majorana-mode localization and transport

1. Majorana-mode localization

Alternating sublattice terminations allow selective control of the upper and lower corner pairs in a bearded hexagonal flake. At $\chi = 0$ and $3|\lambda|/2 < |\Delta_s| < 3|\lambda|$, Eq. (7) gives negative effective mass on every oblique edge. A positive mass on either horizontal edge therefore produces a Majorana mode at each of its endpoints. The upper and lower edges terminate on the B and A sublattices, with effective onsite potentials $-M-\mu$ and $M-\mu$, respectively. Their masses are positive when

$$\begin{aligned} (M+\mu)^2 + |\Delta_s|^2 < 9\lambda^2 \qquad \text{(upper)},\\ (M-\mu)^2 + |\Delta_s|^2 < 9\lambda^2 \qquad \text{(lower)}. \end{aligned} \tag{B1}$$

These inequalities define the overlapping strips in Fig. B1(a) within the leading-order edge theory. Their overlap hosts four Majorana corner modes; within either strip outside the overlap, only the corresponding pair remains. Outside both strips, all edge masses are negative and no mass domain walls occur.

Along $\mu = -M$, the effective onsite potentials vanish on the upper edge and equal $2M$ on the lower edge. Equation (B1) thus retains only the upper pair when $4M^2 + |\Delta_s|^2 > 9\lambda^2$, as illustrated by the finite-flake result in Fig. B1(b). Along $\mu = M$, the upper and lower pairs exchange roles [Fig. B1(d)]. At $M = 0$, the two thresholds coincide, so tuning μ removes both pairs simultaneously within the edge theory.

2. Majorana-mode transport on a hexagonal flake

Varying χ rotates the orbital altermagnetic texture while keeping the Néel vector fixed. The resulting edge-mass reversals move the domain walls, with the gap-closing condition on edge j given by

$$3\left|\lambda\cos\left[2(\theta_j-\chi)\right]\right| = \sqrt{(\xi_j M-\mu)^2+|\Delta_s|^2}. \tag{B2}$$

When the neighboring edges have opposite mass signs, reversing the intervening edge mass transfers one domain wall between its endpoints. The mode spreads along that edge during the transition and then relocalizes at the adjacent corner.

For $M = \mu = 0$, opposite edges share the effective mass $m_{\mathrm{edge},j} = 3|\lambda\cos[2(\theta_j-\chi)]| - |\Delta_s|$ and reverse their signs simultaneously. Among the three edge orientations, the smallest altermagnetic mass magnitude never exceeds $3|\lambda|/2$, whereas the largest is at least $3\sqrt{3}|\lambda|/2$. Thus, positive and negative effective masses coexist at every noncritical orientation when

$$\frac{3}{2}|\lambda| < |\Delta_s| < \frac{3\sqrt{3}}{2}|\lambda|. \tag{B3}$$

The repeated mass pattern on opposite edges then produces four domain walls, whose motion proceeds through paired transfers.

Figure B2 illustrates this mechanism for $\lambda = 0.08t$ and $|\Delta_s| = 0.15t$. Define $\alpha = \frac{1}{2}\arccos[|\Delta_s|/(3|\lambda|)]$, which satisfies $\pi/12 < \alpha < \pi/6$. Equation (B2) gives the first three mass reversals at $\chi = \pi/6-\alpha$, α, and $\pi/3-\alpha$, respectively. Starting from the four horizontal endpoints [Fig. B2(a)], γ_2 and γ_4 move to the lateral vertices when the upper-right and lower-left edges reverse their masses [Fig. B2(b)]. The horizontal edges reverse next, transferring γ_1 and γ_3 to their opposite endpoints [Fig. B2(c)].

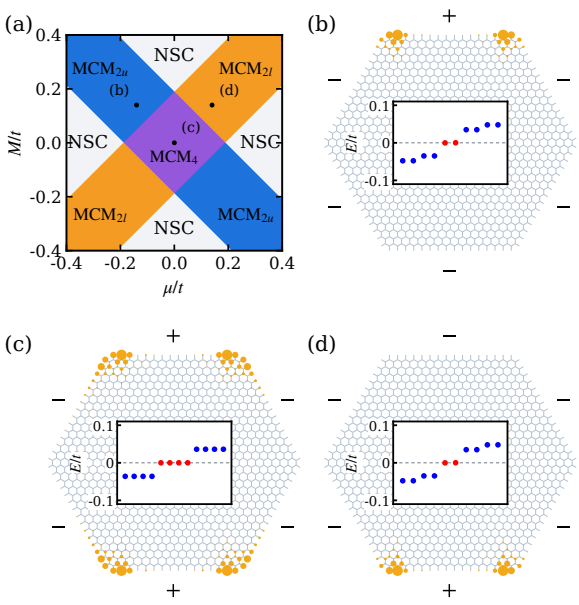


FIG. B1. (a) Phase diagram for the bearded hexagonal flake. NSC denotes the normal-superconductor phase with no Majorana corner modes. MCM_{2u} denotes the second-order topological superconducting phase hosting two Majorana corner modes localized at the ends of the upper horizontal edge. MCM_4 denotes the second-order topological phase supporting four Majorana corner modes. MCM_{2l} denotes the second-order topological superconducting phase hosting two Majorana corner modes localized at the ends of the lower horizontal edge. Phase boundaries agree with the horizontal-edge criterion given in Eq. (B1). (b)–(d) present density distributions of the upper Majorana corner pair, four Majorana corner modes, and lower Majorana corner pair at $(M/t, \mu/t) = (0.14, -0.14)$, $(0, 0)$, and $(0.14, 0.14)$. These parameters are highlighted by black points in (a). Insets show the low-energy spectrum. Red dots mark the Majorana corner modes. Common parameters are at $\chi = 0$, $t' = 0.03t$, $\lambda = 0.08t$, and $|\Delta_s| = 0.15t$.

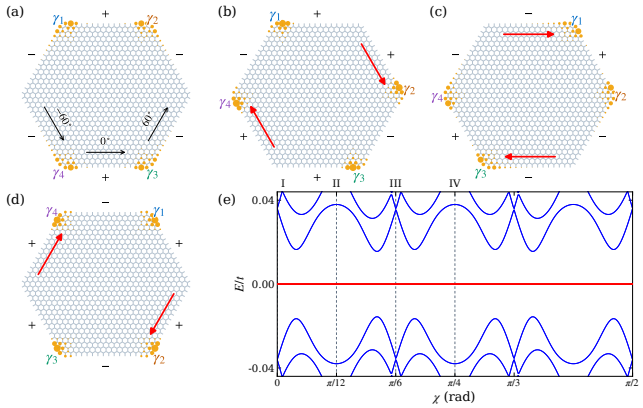


FIG. B2. Paired corner-mode transport in a bearded hexagon at $M = \mu = 0$. (a)–(d) Summed site probabilities of four Majorana modes at $\chi = 0, \pi/12, \pi/6, \pi/4$, with a common marker-size scale. Colored labels track the modes; arrows show schematic transfer directions. The + and − signs in (a)–(d) indicate the signs of the effective edge masses $m_{\mathrm{edge},j}$ at the corresponding values of χ. (e) Low-energy BdG spectrum. Red curves denote the four nearest-zero levels and blue curves the other displayed levels. Guides I–IV correspond to (a)–(d). The hexagon contains 1502 sites, with $\lambda = 0.08t$, $|\Delta_s| = 0.15t$, and $t' = 0.03t$.

The remaining oblique edges then reverse, carrying γ_2 and γ_4 to the adjacent horizontal endpoints. By $\chi = \pi/4$, the original four corners are occupied again, with the mode labels cyclically permuted [Fig. B2(d)].

The crossing angles above are predictions of the leading-order edge theory. The finite-flake spectrum in Fig. B2(e) shows four near-zero-energy states separated from the other displayed levels at the sampled angles. The effective edge-mass pattern has period $\pi/2$, whereas the Hamiltonian with fixed Néel vector has period π. Consequently, the return to the original set of occupied corners at $\chi = \pi/4$ represents a spatial permutation, not a closed Hamiltonian cycle.

Appendix C: Geometric evolution and finite-size effects

1. Geometric evolution and logical gates

The geometric action of the braiding paths in Sec. IV is obtained by transporting a Majorana frame within the isolated low-energy subspace. Let $P(s)$ project onto a particle–hole-invariant $2N$-dimensional BdG subspace along the parameter curve $\mathbf{p}(s)$, where $s \in [0,1]$ and the physical time is Ts. We choose an orthonormal frame satisfying $\mathcal{C}\psi_a = \psi_a$; its vectors need not be energy eigenstates. The corresponding Majorana operators are $\gamma_a = \sqrt{2}\psi_a^\dagger \hat{\Psi}$, where $\hat{\Psi}$ is the real-space Nambu field, and satisfy $\{\gamma_a, \gamma_b\} = 2\delta_{ab}$. The connection $\mathcal{A}_{ab}(s) = \langle \psi_a(s) | \partial_s \psi_b(s) \rangle$ is real and antisymmetric.

The single-flake Majorana pair corresponds to $N = 1$. The X and H gates use four-mode subspaces ($N = 2$), whereas the CNOT gate uses a six-mode subspace ($N = 3$). Writing $\Phi = (\psi_1, \ldots, \psi_{2N})$, the parallel-transported frame is $\Phi_{\mathrm{PT}} = \Phi W_{\mathrm{PT}}$, with $\partial_s W_{\mathrm{PT}} = -\mathcal{A} W_{\mathrm{PT}}$ and $W_{\mathrm{PT}}(0) = I_{2N}$. Endpoint matching for a closed parameter cycle gives

$$W = O_{\mathrm{end}} \mathcal{P} \exp\left[-\int_0^1 \mathcal{A}(s)\, \mathrm{d}s \right], \tag{C1}$$

where $O_{\mathrm{end},ab} = \langle \psi_a(0) | \psi_b(1) \rangle$ and $\mathcal{P}$ denotes path ordering. For parity-preserving transport, $W \in \mathrm{SO}(2N)$.

Numerically, let F_k be the aligned frame at the kth sampling point and Φ_{k+1} the frame at the next point. We form the real overlap matrix $O_k = \mathrm{Re}(\Phi_{k+1}^\dagger F_k)$ and take its singular value decomposition, $O_k = L_k \Sigma_k R_k^{\mathsf{T}}$. The next aligned frame is $F_{k+1} = \Phi_{k+1} L_k R_k^{\mathsf{T}}$. For a closed path, the transport matrix is then $W = \mathrm{Re}(F_0^\dagger F_{\mathrm{end}})$.

For the single triangular flake, a spin rotation of π around the z axis relates the Hamiltonians at χ and $\chi + \pi/2$, which therefore have identical spectra. With the Néel vector fixed, the full parameter cycle is $\chi : 0 \to \pi$. Endpoint matching along this cycle gives $W = -I_2$, consistent with Eq. (13).

The associated many-body unitary operator $\hat{U}_W$ is defined in the initial frame by $\hat{U}_W \gamma_a \hat{U}_W^\dagger = \sum_b W_{ba} \gamma_b$. For an oriented Majorana exchange, we write $U_{ab}(\eta) = \exp(\eta \pi \gamma_a \gamma_b / 4)$, where $\eta = \pm 1$ fixes the exchange orientation. It satisfies $U_{ab} \gamma_a U_{ab}^\dagger = -\eta \gamma_b$ and $U_{ab} \gamma_b U_{ab}^\dagger = \eta \gamma_a$. All exchange operators are expressed in the common initial reference frame, with the rightmost factor acting first. Their indices refer to the initial Majorana labels.

Let K be a real antisymmetric matrix such that $W = e^K$. Its many-body lift reads

$$\hat{U}_W = \exp\left(\frac{1}{4} \sum_{a,b} K_{ab} \gamma_a \gamma_b \right). \tag{C2}$$

The two possible lifts differ by an overall sign. The initial Majorana labels and the ordering of creation operators follow Sec. IV. We restrict $\hat{U}_W$ to the logical basis $\{|0\rangle_{\mathrm{L}}, |1\rangle_{\mathrm{L}}\}$ for the single-qubit gates and $\{|00\rangle_{\mathrm{L}}, |01\rangle_{\mathrm{L}}, |10\rangle_{\mathrm{L}}, |11\rangle_{\mathrm{L}}\}$ for CNOT. The deviation of the resulting geometric operator U_G^{geom} from the ideal gate G is

$$\delta_G = \min_{\alpha \in \mathbb{R}} \left\| U_G^{\mathrm{geom}} - e^{i\alpha} G \right\|_F, \tag{C3}$$

where α is a real global phase angle, $G = X, H, \mathrm{CNOT}$, and $\|\cdot\|_F$ denotes the Frobenius norm. The minimization removes the overall phase while retaining relative logical phases. Thus, δ_G measures the geometric gate deviation, not finite-time infidelity.

For the X gate, the transport matrix of the closed exchange is applied twice. The H gate is evaluated along the full three-exchange path. For CNOT, the five oriented exchanges in Eq. (20) give $W_{\mathrm{CNOT}} =$

TABLE I. Spectral and geometric checks for the quantum gates. Extrema are taken over the sampled control paths, with local refinement for CNOT. The geometric distance δ_G is defined in Eq. (C3).

Gate	$\Delta_{\rm ex}/t$	$\varepsilon_{\rm split}/t$	$r_{\max}$	δ_G
X	3.91×10^{-3}	1.84×10^{-4}	3.43×10^{-2}	5.65×10^{-3}
H	1.54×10^{-3}	6.37×10^{-7}	5.81×10^{-5}	2.04×10^{-5}
CNOT	1.64×10^{-3}	5.69×10^{-6}	3.09×10^{-4}	1.87×10^{-4}

$W_5W_4W_3W_2W_1$, where W_k is the kth exchange matrix in chronological order. These matrices are evaluated in the common initial six-mode frame; reversing a closed exchange replaces its matrix by its transpose. The control segments connect continuously and return to the initial control configuration.

The initial Hadamard parameters, in island order (A,B,C), are $(\chi_A,\chi_B,\chi_C) = (0,\pi/4,\pi/4)$, $(\mu_A,\mu_B,\mu_C) = (0.32,0.08,0.08)t$, and $M_a = 0$. For CNOT, they are $\chi_A = \chi_B = \pi/12$, $\chi_C = \chi_D = \pi/30$, $(\mu_C,\mu_A,\mu_B,\mu_D) = (0.21,-0.05,0.05,0.11)t$, and $M_a = 0.05t$. The control paths are shown in Figs. 6(c) and 7(c).

The elementary closed exchange used for X is sampled adaptively, with 148 accepted points. For H, the complete path contains 2011 points. Its transport matrix differs from the coarser three-exchange composition by 8.93×10^{-8} in Frobenius norm. For CNOT, refining the sampling of the constituent exchanges changes the composite transport matrix by 3.51×10^{-6}.

2. Finite-size gaps and adiabatic conditions

Let $e_{\rm in}(s)$ be the largest $|E|$ inside $P(s)$ and $e_{\rm out}(s)$ the smallest $|E|$ outside it. The spectral separation along the path is $\Delta_{\rm ex} = \min_s[e_{\rm out}(s) - e_{\rm in}(s)]$, and the maximum local splitting ratio is $r_{\max} = \max_s[e_{\rm in}(s)/e_{\rm out}(s)]$.

Delocalization along an edge can increase the hybridization between Majorana modes without closing the spectral separation from the other states. Up to an additive constant, the low-energy Hamiltonian takes the form

$$H_{\rm low} = \frac{i}{2}\sum_{a<b}\varepsilon_{ab}\gamma_a\gamma_b + \sum_n E_n d_n^\dagger d_n, \tag{C4}$$

where ε is real and antisymmetric, and d_n annihilates an additional quasiparticle excitation with energy $E_n > 0$. With this normalization, $\|\varepsilon(s)\|_2 = e_{\rm in}(s)$. The maximum internal splitting is therefore $\varepsilon_{\rm split} = \max_s \|\varepsilon(s)\|_2 = \max_s e_{\rm in}(s)$.

Table I lists these spectral quantities and the geometric gate distances. The CNOT path is locally refined near the spectral bottlenecks. All extrema are evaluated on the sampled paths and do not constitute rigorous bounds along the continuous curves.

A favorable operating hierarchy is $\varepsilon_{\rm split} \ll \hbar/T \ll \Delta_{\rm ex}$ [68]. The first inequality limits the accumulation of internal dynamical phases [69]. The ratios $\varepsilon_{\rm split}/\Delta_{\rm ex}$ are 4.71×10^{-2}, 4.15×10^{-4}, and 3.47×10^{-3} for X, H, and CNOT, respectively. The separation of these energy scales is therefore more limited for X.

Suppression of transitions out of the low-energy subspace also depends on the control path and sweep speed [70]. For physical time Ts, the local adiabaticity condition reads

$$\frac{\hbar}{T}\max_{s,a,n}\frac{|\langle n|\partial_s H_{\rm BdG}|a\rangle|}{|E_n - E_a|^2} \ll 1, \tag{C5}$$

where $|a\rangle$ and $|n\rangle$ are instantaneous BdG eigenstates inside and outside $P(s)$, respectively. The spectral hierarchy alone does not establish this condition; its evaluation requires the transition matrix elements along the chosen path.

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